

Content Injection (T1659)

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1 Attack Description in CNL

The attack description of the MITRE ATT&CK Technique “*Content Injection*” (T1659)¹ written in the proposed CNL is shown below.

```
1 Tactics: TA0001 Initial_Access, TA0011 Command_and_Control
2 Technique: T1659 Content_Injection
3
4 Background
5 Given process as browser
6 And endpoint as server
7 And network_connection as Browser_connection with source = "browser", destination = "server"
8 And process as local_process with parent_process = "browser"
9 And file as F
10 And directory as Temp with path = "/temp/"
11 And directory as AppData with path = "/%AppData%/"
12
13 Event 1
14 Given Browser_connection is activated
15 When browser receives F from server
16 Then F is received
17
18 Event 2
19 Given F is received
20 When browser saves F in AppData
21 Then F exists in AppData
22
23 Event 3
24 Given F is received
25 When browser saves F in Temp
26 Then F exists in Temp
27
28 Event 4
29 Given F exists in AppData Xor F exists in Temp
30 When local_process executes F
31 Then F is active
32
33 Completion
34 Event 4
```

¹See <https://attack.mitre.org/techniques/T1659>

2 Attack Model (Petri Net)

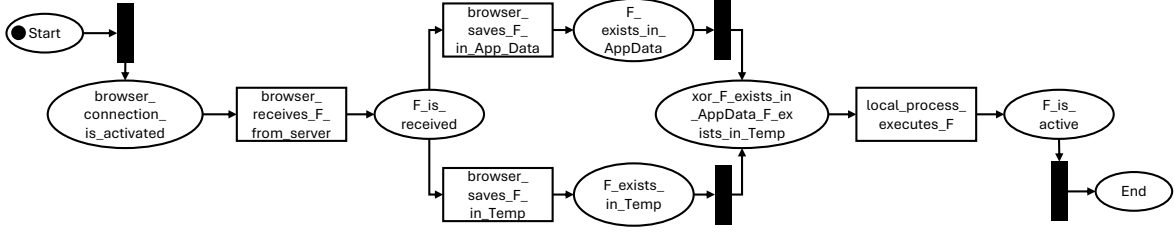


Figure 1: Content Injection Petri Net

3 Runtime and Memory Usage

The translation of this CNL description took 514ms and allocated 606 KB at peak. The translation pipeline was run on a laptop with Intel Core i7-12650H, 16 GB RAM and running Windows 11.

4 Validation Criteria

The validation process included semi-structured interviews with two domain experts, in which the following validation criteria were discussed: **V1.** the technique description (in CNL) expresses the content of the ATT&CK text; **V2.** the technique description (in CNL) does not contain ambiguities; **V3.** the Technique Model captures the described attack features; **V4.** the Petri Net presents the alternative paths the attack can follow based on the described actions, pre- and post-conditions and assets involved. In the discussion section below, the domain experts are referred to as E1 and E2.

5 Discussion

5.1 V1 & V2

The CNL description for *Content Injection* contains the attack features present in the ATT&CK text and matches the high level of abstraction that they are originally described in. The CNL captures the wide variety of assets involved in the attack, and allows for their explicit compositional declaration. For example, *Browser connection*, through which a malicious file can enter the system, is a complex asset type, that contains other assets in its properties. The CNL description expresses this compositionality explicitly and fully. The event statements also, are specific regarding the alternative locations in which a received file *F* could be stored in during the attack. The language enables this expression while maintaining the description short and avoiding duplicate expression of future actions, with the use of *Xor* in the preconditions of Event 4. Overall, assets are described as stated in the source description, with their relation to other asset clearly defined, and without introducing new ambiguities. During the interview, E2, in particular, also stated that the CNL description expresses the assets well and in alignment with the source.

5.2 V3 & V4

The produced Technique Model effectively captures the attack features as well as the complex compositionality of the participating assets in their appropriate types and names. Similarly, the Petri Net showcases the system behavior in relation to the attack, mapping accurately the events to transitions and the pre- and post-conditions to intermediate places. It also clearly shows the alternative paths the attack can follow and which assets are involved in each step. During the interview, E2 highlighted the valuable insights they can extract from the CNL description and the produced Petri Net. These included the vulnerable system assets that are participating in each attack step, and the procedural information of the attack’s execution that allow them to better prepare and defend their system against the attack.